

FluxLock Protocol (FLCP)

A Post-Quantum, State-Evolving Cryptographic Network

Abstract

FluxLock introduces a cryptographic protocol designed for a post-quantum world. Rather than relying on static public/private key pairs, FluxLock defines ownership as the ability to continuously evolve a cryptographic state synchronized with network entropy. This paper presents the motivation, formal model, security assumptions, and a novel consensus mechanism built around observer-locked validation and forward-only state transitions.

1. Motivation

Quantum computing threatens all cryptographic systems based on long-lived secrets. Even post-quantum algorithms largely preserve the paradigm of static identity. FluxLock rejects this assumption entirely, proposing cryptography based on motion, synchronization, and irreversible state evolution.

2. Core Concept: State-Based Identity

In FluxLock, a wallet is not a keypair but a state machine. Each valid action advances the wallet into a new cryptographic state. Past states are immediately invalidated and cannot be reused or reconstructed.

State transition function: $S_{\text{new}} = H(S_{\text{old}} \parallel T \parallel E \parallel N)$

3. Flux Signatures

Authorization is achieved through Flux Signatures. A transaction proves knowledge of the previous state while demonstrating synchronization with time and network entropy. No static secret is ever transmitted or stored.

4. Security Model

FluxLock provides forward security, replay resistance, and quantum attack mitigation. Captured states expire instantly. Even future cryptographic breaks cannot retroactively compromise past transactions.

5. Consensus: Observer-Locked Proof of Continuity (OLPC)

FluxLock employs Observer-Locked Proof of Continuity (OLPC). Validators act as temporal witnesses, co-signing valid state transitions within strict time windows. Consensus emerges from agreement on the correct ordering of state evolutions rather than competition for hash power.

Validators contribute entropy, verify state transitions, and attest to temporal validity. An attacker cannot precompute or replay transactions without live participation in the network.

6. Recovery and Continuity

Recovery is handled through State Anchors rather than seed phrases. Trusted quorums can reconstitute a wallet's latest valid state without revealing historical data.

7. Conclusion

FluxLock reframes cryptography around continuous synchronization. By eliminating static secrets and embracing irreversible state evolution, it offers a path toward resilient, post-quantum decentralized systems.